CHAPTER 5: APPLICATION LAYER AND DOMAIN NAME SYSTEM (DNS)

5.1 The Domain Name System (DNS)

Humans remember mnemonic host names such as 'university.edu' or 'wikipedia.org', while IP routers require numerical IP addresses (e.g., 192.0.2.1 for IPv4 or 2001:db8::1 for IPv6). The Domain Name System provides a globally distributed, hierarchical database that translates human-friendly host names into numerical IP addresses.

5.2 DNS Resolution Hierarchy

The DNS namespace is organized into a tree hierarchy:

1. Root DNS Servers: Top of the tree; contact Top-Level Domain (TLD) servers.
2. TLD Servers: Manage domains such as .com, .org, .edu, and country codes like .in or .uk.
3. Authoritative DNS Servers: Host the actual DNS resource records for specific organizations.
4. Local DNS Resolver / Cache: Caches mappings locally to speed up subsequent queries and reduce network round-trips.

5.3 Transport Layer: TCP vs UDP

- Transmission Control Protocol (TCP): Connection-oriented, reliable protocol providing flow control, congestion control, and ordered segment delivery via a 3-way handshake (SYN, SYN-ACK, ACK).
- User Datagram Protocol (UDP): Connectionless, lightweight transport protocol with no guaranteed delivery, minimal overhead, and low latency, widely used for DNS queries, video streaming, and VoIP.